

Successful completion of the course, the students will be able to:

- 01: Learn about the sand casting process.
 02: Understand the special casting process and casting defects.
 03: Learn about different metal forming processes.
 04: Understand different metal joining processes and their characteristics.
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 O – Programme Outcome

PART A

(Answer **ALL** questions)

		(8 × 3 = 24)	Marks	BL
I.				
(a)	In sand casting, why the core is made stronger than cavity part? What are the forces acting on the core?		3	L2
(b)	State different types of melting furnaces and their heat generation principle, Which are used in melting of casting metals?		3	L1
(c)	Explain the process of Slush casting.		3	L1
(d)	What are the design considerations to be made while designing riser in a sand casting?		3	L4
(e)	Differentiate between Isotropic hardening and Kinematic hardening.		3	L2
(f)	Explain the Von-Mises criteria of metal yielding.		3	L1
(g)	List the advantages of DC welding machine over AC welding machine.		3	L2
(h)	Explain the process of Thermit welding.		3	L1

PART B

(4 × 12 = 48)

II.	(a)	A rectangular shaped metal part has to be made by sand casting. The dimensions of the component is given as: length 100 mm, width 50 mm and thickness 25 mm. Find the dimensions of wooden pattern which is required to cast this component. The shrinkage allowance is given as 0.03mm/ mm length.	6	L5
	(b)	If the casting mentioned in the question (a) solidifies in 20 minutes, find the solidification time for another casting of same metal with dimensions 70 mm × 50 mm × 15 mm, under similar conditions.	6	L4

OR

III.	(a)	Explain different types of patterns with figure.	8	L1
	(b)	What are the different types of core supporting methods which are used in sand casting process? Explain with figure.	4	L1

T T D

- IV. (a) What are the design aspects which can make the casting components economical? 6 L3 2 1,2
 (b) A cube of side 'a' and a solid cylinder of diameter 'd' are made from same metal. Find the ratio of their solidification time as per Chvorinov's rule. 6 L4 2 1,2

OR

- V. (a) Explain with figures, the process of wax casting. 7 L2 2 1,2
 (b) Compare the advantages and disadvantages of hot chamber and cold chamber die casting processes. 5 L2 2 1,2

- VI. (a) For a single pass rolling process, the initial thickness of the strip is 3 mm, final thickness is 2.5 mm, roll diameter is 250 mm, Width of the strip is 450 mm, average friction coefficient is 0.8, and plain flow stress is 14 MPa. Find the angle of bite, Roll contact length, Roll separating force, and Torque per roll. 8 L5 3 1,2
 (b) Differentiate between direct extrusion and indirect extrusion with figure. 4 L2 3 1,2

OR

- VII. (a) A hollow washer of inner diameter 25mm and outer diameter 100 mm is to be made from a cold rolled steel of 3 mm thickness. Find the diameter of punch and die for both punching and blanking process. Take the cutting clearance as 0.1 mm. Also find the blanking and punching force required, if the shear strength of the steel is 300 MPa. 8 L3 3 1,2
 (b) Differentiate between hot rolling and cold rolling operations. 4 L2 3 1,2

- VIII. (a) What do you mean by duty cycle of welding? Find the duty cycle of the welding process when heat generation is 12 kW, heat dissipation is 7 kW, maximum heat accumulation allowed is 2000 KJ. 6 L5 4 1,2
 (b) A direct current welding machine with linear power source characteristics provides open circuit voltage of 80 V and short circuit current of 800 A. During welding, the arc current is 500A while the arc length is 5mm, and arc current is 460A with arc length 7mm. Find the linear relation between voltage and arc length. 6 L3 4 1,2

OR

- IX. (a) What are the limitations and advantages of Submerged arc welding? 4 L2 4 1,2
 (b) Differentiate between soldering and brazing process. 4 L2 4 1,2
 (c) Explain the working principle of Electron beam welding. 4 L1 4 1,2

Blooms's Taxonomy Levels

L1 - 24%, L2 - 30%, L3 - 16%, L4 - 13%, L5 - 17%, L6 - 0%,

ADVANTAGE

ACCURATE

SURFACE FINISH

ARC CANT SEE

EXPENSIVE

AUTOMATIC